

Week 5: Healthcare Data Cleaning and Data Modeling

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1. Introduction

The primary objective of Week 5 focuses on executing Healthcare Data Cleaning and Data Modeling within Power BI. The project began by taking the raw healthcare dataset and loading it into Power Query to clean missing values, eliminate data inconsistencies, and normalize the flat file into separate dimension and fact tables. Following data prep, a normalized Star Schema Data Model was established by connecting Dim_Patient and Dim_Admission dimension tables to the central Fact_Billing table. Finally, a dedicated Measures Table was created to house analytical business logic, where core Key Performance Indicators (KPIs) were computed using Data Analysis Expressions (DAX).

2. Data Cleaning & Transformation in Power Query

To prepare the raw healthcare dataset for analytical processing, fundamental cleaning and standardization steps were performed in Power Query. Text fields (such as Name and Medical Condition) were standardized using proper casing, while explicit data types were assigned across numerical and date attributes to ensure data consistency and quality.

Queries [1] Table.TransformColumns(#"Changed Type",{{"Name", Text.Proper, type text}})

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	Bobby Jackson	30	Male	B-	Cancer	31-01-2024	Matthew Sml
2	Leslie Terry	62	Male	A+	Obesity	20-08-2019	Samantha Da
3	Danny Smith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitch
4	Andrew Watts	28	Female	O+	Diabetes	18-11-2020	Kevin Wells
5	Adrienne Bell	43	Female	AB+	Cancer	19-09-2022	Kathleen Han
6	Emily Johnson	36	Male	A+	Asthma	20-12-2023	Taylor Nevito
7	Edward Edwards	21	Female	AB-	Diabetes	03-11-2020	Kelly Olson
8	Christina Martinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thor
9	Jasmine Aguilar	82	Male	AB+	Asthma	01-07-2020	Daniel Fergus
10	Christopher Berg	38	Female	AB-	Cancer	23-05-2021	Heather Day
11	Michelle Daniels	72	Male	O+	Cancer	19-04-2020	John Duncan
12	Aaron Martinez	38	Female	A-	Hypertension	13-08-2023	Douglas May
13	Connor Hansen	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletc
14	Robert Bauer	68	Female	AB+	Asthma	22-05-2020	Theresa Freee
15	Brooke Brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stew
16	Ms. Natalie Gamble	46	Female	AB-	Obesity	01-01-2023	Maria Dough
17	Haley Perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer
18	Mrs. Jamie Campbell	38	Male	AB-	Obesity	08-03-2020	Justin Kim
19	Luke Burgess	24	Female	A-	Hypertension	04-03-2021	Justin Moore
20	Daniel Schmidt	63	Male	B+	Asthma	15-11-2022	Denise Gallov
21	Timothy Burns	67	Female	A-	Asthma	28-06-2023	Krista Smith
22	Christopher Bright	48	Male	B+	Asthma	21-01-2020	Gregory Smit
23	Kathryn Stewart	58	Female	O+	Arthritis	12-05-2022	Vanessa New
24	Dr. Eileen Thompson	59	Male	A+	Asthma	02-08-2021	Donna Martir
25	Paul Henderson	72	Female	AB+	Hypertension	15-05-2020	Stephanie Kru
26	Peter Fitzgerald	73	Male	AB+	Obesity	15-05-2020	Angela Contr
27	Cathy Small	51	Female	O-	Asthma	23-12-2023	Wendy Glenn
28	Mr. Kenneth Moore	34	Female	A+	Diabetes	21-06-2022	James Ellis
29	Mary Hunter	38	Female	O-	Hypertension	03-01-2021	Jared Bruce Ji

3. Query Merging & Key Alignment

To construct the Star Schema, entity records were aligned across tables using key mapping techniques. A Left Outer Join was executed between the transactional billing data and Dim_Patient to establish surrogate key connections, ensuring complete record matching and relational integrity.

Discharge Date

100%	Valid	100%
0%	Error	0%
0%	Empty	0%

1 02-02-2022

2 26-08-2021

3 07-10-2022

4 18-12-2022

5 09-10-2022

6 24-12-2022

7 15-11-2022

8 07-01-2022

9 14-07-2022

10 22-06-2022

11 22-04-2022

12 05-09-2022

13 28-12-2021

14 19-06-2022

15 13-10-2022

16 11-01-2022

17 14-07-2022

18 02-04-2022

19 14-03-2022

20 22-11-2022

21 02-07-2022

22 09-02-2022

23 10-06-2022

24 12-08-2022

25 08-06-2022

26 20-05-2022

27 19-01-2024 Ibuprofen Normal 30-60 Cathy Small 51|Female|O-

28 30-06-2022 Lipitor Abnormal 19-35 Mr. Kenneth Moore 34|Female|A+

29 16-01-2021 Penicillin Normal 30-60 Mary Hunter 38|Female|O-

30

Merge

Select a table and matching columns to create a merged table.

Fact_Billing

Name	1	Age	2	Gender	3	Blood Type	Medical Condition	Date of Admission	Doctor
Bobby Jackson		30		Male		B-	Cancer	31-01-2024	Matthew Smith
Leslie Terry		62		Male		A+	Obesity	20-08-2019	Samantha Davies
Danny Smith		76		Female		A-	Obesity	22-09-2022	Tiffany Mitchell
Andrew Watts		28		Female		O+	Diabetes	18-11-2020	Kevin Wells

Dim_Patient

Name	1	Age	2	Gender	3	Blood Type	Age Group	Patient ID
Bobby Jackson		30		Male		B-	19-35	1
Leslie Terry		62		Male		A+	Above 60	2
Danny Smith		76		Female		A-	Above 60	3
Andrew Watts		28		Female		O+	19-35	4
Adrienne Bell		43		Female		AB+	30-60	5

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

✓ The selection matches 40735 of 55500 rows from the first table.

OK Cancel

Admission ID

Valid	100%
Error	0%
Empty	0%

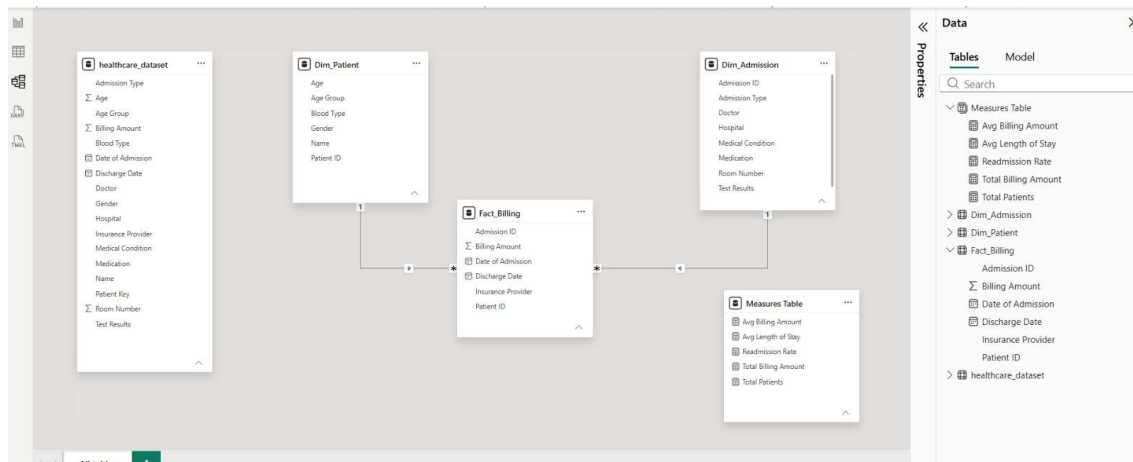
4. Fact Table Optimization

The core transactional entity, Fact_Billing, was refined to store essential foreign keys (Patient ID, Admission ID), transaction dates (Date of Admission, Discharge Date), insurance attributes, and primary numerical metrics (Billing Amount) necessary for aggregation.

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Admission ID	Patient ID
	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 100%, Error 0%, Empty 0%	Valid 99%, Error 0%, Empty < 1%
1	31-01-2024	Blue Cross	18856.28131	02-02-2024	1	1
2	20-08-2019	Medicare	33643.32729	26-08-2019	2	2
3	22-09-2022	Aetna	27955.09608	07-10-2022	3	3
4	18-11-2020	Medicare	37909.78241	18-12-2020	4	4
5	19-09-2022	Aetna	14238.31781	09-10-2022	5	5
6	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023	6	6
7	03-11-2020	Medicare	19580.87234	15-11-2020	7	7
8	28-12-2021	Cigna	45820.46272	07-01-2022	8	8
9	01-07-2020	Cigna	50119.22279	14-07-2020	9	9
10	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021	10	10
11	19-04-2020	Medicare	12576.79561	22-04-2020	11	11
12	13-08-2023	Medicare	7999.58688	05-09-2023	12	12
13	12-12-2019	Cigna	43282.28336	28-12-2019	13	13
14	22-05-2020	UnitedHealthcare	33207.70663	19-06-2020	14	14
15	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021	15	15
16	01-01-2023	Blue Cross	12263.35743	11-01-2023	16	16
17	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020	17	17
18	08-03-2020	Cigna	17440.46544	02-04-2020	18	18
19	04-03-2021	Blue Cross	18843.02302	14-03-2021	19	19
20	15-11-2022	Cigna	23762.20358	22-11-2022	20	20
21	28-06-2023	Blue Cross	42.51458855	02-07-2023	21	21
22	21-01-2020	Aetna	17695.91162	09-02-2020	22	22
23	12-05-2022	Aetna	5998.102908	10-06-2022	23	23
24	02-08-2021	Aetna	25250.05243	12-08-2021	24	24
25	15-05-2020	Medicare	33211.29542	08-06-2020	25	25
26	15-05-2020	Medicare	19746.83201	20-05-2020	26	26
27	23-12-2023	Blue Cross	26786.52956	19-01-2024	27	27
28	21-06-2022	UnitedHealthcare	18834.80134	30-06-2022	28	28
29	03-01-2021	Cigna	32643.29935	16-01-2021	29	29
30	03-10-2023	Aetna	6767.011064	13-10-2023	30	30

5. Star Schema Data Modeling Setup

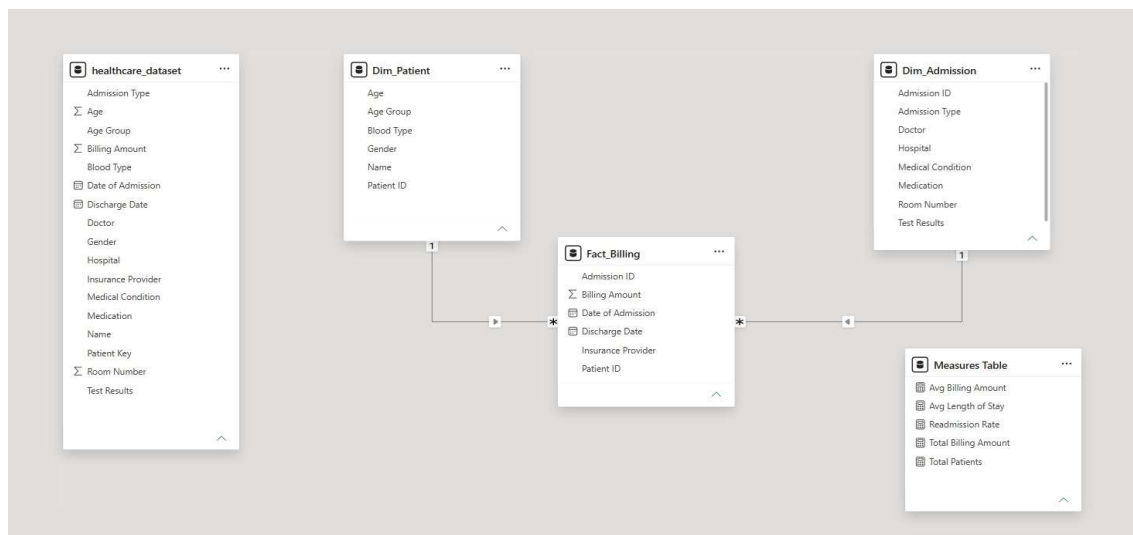
Within the Power BI Model View, a formal Star Schema architecture was established. Dimension tables (Dim_Patient and Dim_Admission) were linked to the central Fact_Billing table using single-directional 1:* (One-to-Many) relationships. An independent table named Measures Table was initialized to serve as a centralized container for all DAX formulas.



6. Finalized Data Model Architecture

The final relationship configuration was validated to guarantee proper filter propagation and model accuracy:

- **Dim_Patient ↔ Fact_Billing**: 1:* relationship mapped on Patient ID
- **Dim_Admission ↔ Fact_Billing**: 1:* relationship mapped on Admission ID
- **Measures Table**: Centralized container housing all KPI measures independently of physical data columns.



7. DAX Implementation & Healthcare Metrics

Following model validation, five primary healthcare metrics were created using Data Analysis Expressions (DAX) and stored within the Measures Table container:

1. Total Patients

Calculates the total number of patient admission records present in the billing dataset.

Total Patients = COUNTROWS(Fact_Billing)

2. Total Healthcare Revenue

Aggregates the total financial revenue generated across all billing records.

Total Billing Amount = SUM(Fact_Billing[Billing Amount])

3. Average Billing Cost per Patient

Computes the mean financial expenditure billed per patient admission.

Avg Billing Amount = AVERAGE(Fact_Billing[Billing Amount])

4. Average Length of Stay (in Days)

Determines the mean duration of hospital stays by calculating the day difference between the admission date and discharge date.

Avg Length of Stay =

AVERAGEX(

Fact_Billing,

DATEDIFF(Fact_Billing[Date of Admission],
Fact_Billing[Discharge Date], DAY)
)

5. Readmission Rate

Calculates the percentage of patients who experienced multiple hospital admissions relative to the overall patient count.

Readmission Rate =

VAR TotalCount = [Total Patients]

VAR RepeatPatients =

```
COUNTROWS(
    FILTER(
        SUMMARIZE(
            Fact_Billing,
            Fact_Billing[Patient ID],
            "AdmissionCount", COUNTROWS(Fact_Billing)
        ),
        [AdmissionCount] > 1
    )
)
RETURN
DIVIDE(RepeatPatients, TotalCount, 0)
```

8. Conclusion

The Week 5 milestone successfully transformed unstructured healthcare records into an optimized, enterprise-grade analytical model. By enforcing strict data cleaning protocols in Power Query and restructuring the data into a normalized Star Schema, relational integrity and optimal query performance were established. The

creation of a dedicated Measures Table housing standardized DAX logic ensures scalable, dynamic calculation of essential healthcare metrics such as patient volume, revenue generation, duration of stay, and patient readmission rates. This robust data modeling foundation fully prepares the dataset for seamless interactive visualization and reporting in subsequent phases of the project.